# PG No. 21 $C_6$ 6 [ hexagonal ]

* generator : $3^+_{001}$, $2_{001}$
* conjugacy class

$[1]$ : 1
$[2_{001}]$ : $2_{001}$
$[3^+_{001}]$ : $3^+_{001}$, $3^-_{001}$
$[6^+_{001}]$ : $6^+_{001}$, $6^-_{001}$

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

| No. | tag | matrix (polar) | det |
|---|---|---|---|
| 1 | 1 | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 2 | $3^+_{001}$ | $\begin{bmatrix} 0 & -1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 3 | $3^-_{001}$ | $\begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 4 | $2_{001}$ | $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 5 | $6^-_{001}$ | $\begin{bmatrix} 0 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 6 | $6^+_{001}$ | $\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |